

OUTPUT1-1

```
gorkem@gorkem:~/Desktop$ gcc --ansi 230104004174_part_1.c -o 230104004174_part_1
gorkem@gorkem:~/Desktop$ ./230104004174_part_1
*****MENU*****
1. Sort and display all individuals by age
2. Sort and display individuals in the branch by age
3. Show individuals with the branch 'SCIENCE'
4. Show computer scientists who are not mathematicians
5. Exit
Choice: 1
```

Name	Surname	Age	Branch1	Branch2
Hypatia		35	MATHEMATICS	PHILOSOPHY
Ada	Lovelace	37	COMPUTER SCIENCE	
Canan	Dagdeviren	38	PHYSICS	MATERIALS SCIENCE
Grace	Hopper	41	COMPUTER SCIENCE	MATHEMATICS
Alan	Turing	42	COMPUTER SCIENCE	MATHEMATICS
John von	Neumann	54	COMPUTER SCIENCE	
Özlem	Tureci	57	MEDICINE	IMMUNOLOGY
Aristo		62	PHILOSOPHY	
Gregor	Mendel	62	GENETICS	MATHEMATICS
Mehmet	Oz	63	MEDICINE	
Dmitri	Mendeleev	63	CHEMISTRY	
Marie	Curie	67	CHEMISTRY	PHYSICS
Janaki	Ammal	67	BOTANY	CYTOGENETICS
Sokrates		70	PHILOSOPHY	
Feza	Gursey	71	PHYSICS	MATHEMATICS
Charles	Darwin	73	GENETICS	GEOLOGY
Albert	Einstein	76	PHYSICS	
Aziz	Sancar	77	CHEMISTRY	MEDICINE
John	Dalton	78	CHEMISTRY	PHYSICS

OUTPUT1-2

Charles	Darwin	73	GENETICS	GEOLOGY
Albert	Einstein	76	PHYSICS	
Aziz	Sancar	77	CHEMISTRY	MEDICINE
John	Dalton	78	CHEMISTRY	PHYSICS
Behram	Kursunoglu	81	PHYSICS	
Isaac	Newton	84	PHYSICS	
Cahit	Arf	87	MATHEMATICS	
Asuman	Baytop	95	BOTANY	PHARMACY

Choice:

OUTPUT1-3

Choice: 2					
Name	Surname	Age	Branch1	Branch2	
Janaki	Ammal	67	BOTANY	CYTOGENETICS	
Asuman	Baytop	95	BOTANY	PHARMACY	
Aziz	Sancar	77	CHEMISTRY	MEDICINE	
Marie	Curie	67	CHEMISTRY	PHYSICS	
John	Dalton	78	CHEMISTRY	PHYSICS	
Dmitri	Mendeleev	63	CHEMISTRY		
Grace	Hopper	41	COMPUTER SCIENCE	MATHEMATICS	
Alan	Turing	42	COMPUTER SCIENCE	MATHEMATICS	
Ada	Lovelace	37	COMPUTER SCIENCE		
John von	Neumann	54	COMPUTER SCIENCE		
Charles	Darwin	73	GENETICS	GEOLOGY	
Gregor	Mendel	62	GENETICS	MATHEMATICS	
Hypatia		35	MATHEMATICS	PHILOSOPHY	
Cahit	Arf	87	MATHEMATICS		
Özlem	Tureci	57	MEDICINE	IMMUNOLOGY	
Mehmet	Oz	63	MEDICINE		
Aristo		62	PHILOSOPHY		
Sokrates		70	PHILOSOPHY		
Canan	Dagdeviren	38	PHYSICS	MATERIALS SCIENCE	
Feza	Gursey	71	PHYSICS	MATHEMATICS	
Albert	Einstein	76	PHYSICS		

OUTPUT1-4

Choice:	Aristo	62	PHILOSOPHY		
	Sokrates	70	PHILOSOPHY		
	Canan	Dagdeviren	38	PHYSICS	MATERIALS SCIENCE
	Feza	Gursey	71	PHYSICS	MATHEMATICS
	Albert	Einstein	76	PHYSICS	
	Behram	Kursunoglu	81	PHYSICS	
	Isaac	Newton	84	PHYSICS	

OUTPUT1-5

```
Choice: 3
      Name      Surname      Age      Branch1      Branch2
      Ada      Lovelace      37      COMPUTER SCIENCE
      John von      Neumann      54      COMPUTER SCIENCE
      Alan      Turing      42      COMPUTER SCIENCE      MATHEMATICS
      Canan      Dagdeviren      38      PHYSICS      MATERIALS SCIENCE
      Grace      Hopper      41      COMPUTER SCIENCE      MATHEMATICS
Choice: 4
      Name      Surname      Age      Branch1      Branch2
      Ada      Lovelace      37      COMPUTER SCIENCE
      John von      Neumann      54      COMPUTER SCIENCE
Choice:
```

OUTPUT1-6

```
Choice: 6
Please enter a valid choice!
Choice: -1
Please enter a valid choice!
Choice: 10
Please enter a valid choice!
Choice: 99
Please enter a valid choice!
Choice: 5
gorkem@gorkem:~/Desktop$
```

OUTPUT2-1

```
gorkem@gorkem:~/Desktop$ gcc --ansi 230104004174_part_2.c -o 230104004174_part_2
gorkem@gorkem:~/Desktop$ ./230104004174_part_2
P1 @ (3,3)
P2 @ (1,30)
gorkem@gorkem:~/Desktop$
```

OUTPUT2-2 (With different file)

```
gorkem@gorkem:~/Desktop$ ./230104004174_part_2
P1 @ (1,3)
P1 @ (5,9)
P2 @ (1,1)
P2 @ (1,6)
P2 @ (1,7)
P2 @ (1,8)
P2 @ (1,31)
gorkem@gorkem:~/Desktop$
```

Youtube Link : <https://youtu.be/3NnNT9s6HIU>